



SCIENCE FOR POLICY BRIEF

Putting knowledge and technology to work: Insights from the Innovation Output Indicator

HIGHLIGHTS

- The EU27 leads Canada, Australia, and China in 2023, but is outperformed by economic partners such as the United States, South Korea and Japan. China exhibited a notable increase in performance and is closing its innovation gap with the EU and the US.
- The EU economy performs well when it comes to exporting knowledge-intensive manufactured goods with a high domestic content, but should improve in terms of employment in knowledge-intensive industries and increase the share of firms that are actively engaged in innovation.
- Sweden, Ireland, Germany and Finland are the best performers among EU Member States. Since 2013, performance has improved the most in Lithuania, Estonia, and Sweden, while it has declined in Germany, Luxembourg, Romania, and Spain.
- The EU must address the innovation gap with its main competitors. To this end, it should increase R&D investment, rebalance its efforts towards advanced technologies, and promote a supportive ecosystem for start-ups and innovation.

EU innovation output

As emphasized by Mario Draghi in his recent report on the Future of European Competitiveness [1], innovation is of utmost importance for reigniting economic growth, fostering a competitive Europe, and securing a sustainable future for its citizens. For this reason, monitoring innovation performance has always been a priority of the European Commission. Since 2013, the European Commission has produced the Innovation Output Indicator (IOI) – a composite indicator offering an output-oriented measure of innovation performance at the country and EU levels.

The IOI tracks the extent to which innovative ideas reach the market, create better jobs, and increase a

country's technological capability. It aggregates four components : 1. patent and trademark counts, 2. employment in knowledge-intensive activities in business industries¹, 3. domestic contribution to the value of knowledge-intensive exports, and 4. innovation-active enterprises.

The IOI is a means to compare the innovation output of the EU27 with that of international economic partners. In 2023, the EU leads a group of world players, including Canada, Australia, and China, (Figure 1), but is outperformed by other competitors, such as the United States, South Korea and Japan [3].

The EU27 recorded a 6% increase in its IOI score between 2013 and 2023. This marks an improvement

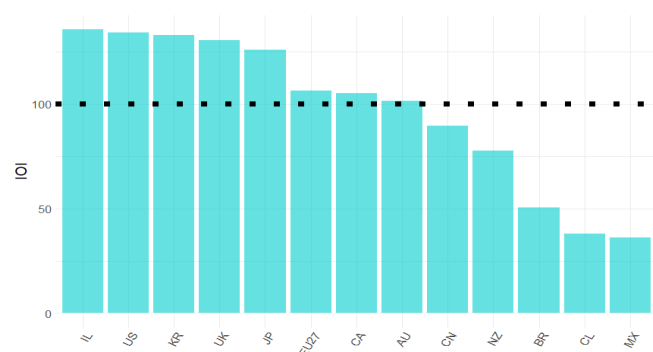
¹ These are NACE manufacturing and services industry sectors with high technological intensity as expressed by their share of tertiary educated employment. See [2] for detailed information.

QUICK GUIDE

Composite indicators and scoreboards are widely used to capture both the process and the outcomes of complex phenomena such as innovation. Since 2013, the Joint Research Centre of the European Commission has produced the Innovation Output Indicator (IOI) within the INNOVA projects. The IOI is an output-oriented measure of innovation for the EU and its Member States that measures countries' capacity to derive economic benefits from innovation. It tracks the extent to which innovative ideas reach the market, create knowledge-intensive jobs, and increase country technological capability.

The IOI includes six indicators combined in four components: 1) the number of Patent Cooperation Treaty (PCT) patent applications, and the number of trademark classes in total resident trademark applications, each denominated by population (in millions); 2) the persons employed in knowledge-intensive activities and business industries as a share of total employment; 3) domestic value added content of medium-high and high-tech manufacturing exports as a share of total manufacturing exports, and domestic value added content of knowledge-intensive service exports as a share of total service exports; and 4) the number of innovation-active firms over the total number of firms.

Figure 1 – Innovation Output Indicator of the EU in 2023, compared to international economic partners



Source: Bello et al. 2025. Note: dotted line shows the EU27 value in 2013.

in innovation output performance, comparable to that of the US (6%), but still less than that reported by Japan (14%). By contrast, China reported a nearly 31% IOI improvement over the same period and appears poised to soon close the innovation gap with the EU. A similar trend is also confirmed by the European Innovation Scoreboard [4]

Sweden, Ireland, Germany and Finland are the top performers among EU Member States (MSs) (Figure 2). Conversely Bulgaria, Latvia, and Romania reported the lowest performance. The IOI results confirm that the innovation divide persists in the EU, with the northern and western Europe being home to innovation leaders [5].

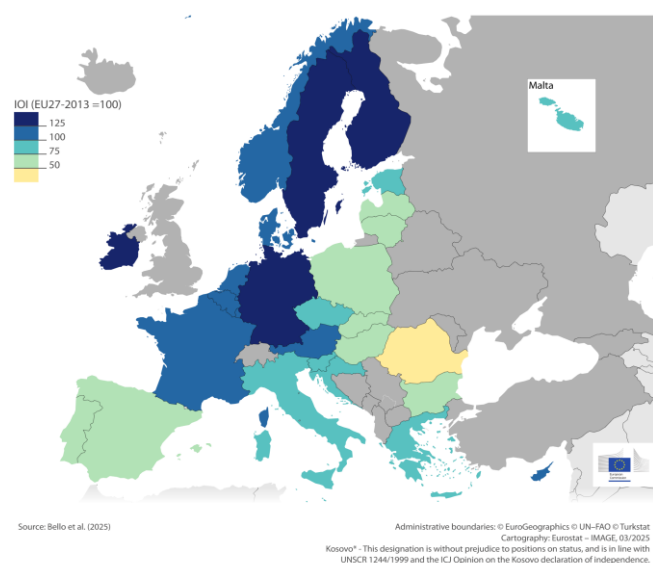
The largest improvements in performance since 2013 took place in countries that are low-ranked in terms of the IOI, such as Lithuania, Bulgaria and Latvia. For Bulgaria, this was driven by improvements in the share of innovative enterprises and knowledge-intensive employment, while Lithuania and Latvia improved the most in terms of patenting activity. Lithuania and Latvia are also among the countries making large improvements in the last years in the WIPO's Global Innovation Index [6], performing well in terms of creative outputs (Latvia) and cumulative value of unicorns (Lithuania).

A closer look at the IOI components

The information contained in the IOI extends beyond a simple country ranking, highlighting significant variation in innovation performance across EU Member States (MS) as measured by the components that make up the index.

According to 2024 IOI results, Sweden has the highest number of Patent Cooperation Treaty (PCT) patent applications per million population (over 300), followed by Denmark and Finland. In contrast, 16 EU MSs have fewer than 30 patents per million population. Malta and Luxembourg lead the EU MS ranking in terms of trademark applications per million population (Figure 3), although they account for less than 1% of the EU27's total trademark applications.

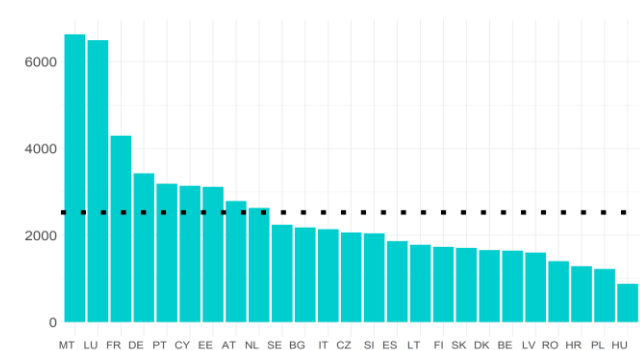
Figure 2 – Innovation Output Indicator of EU Member States, 2023



Source: Bello et al. 2025. Note: values normalized to the EU27 value in 2013 (= 100). Darker colours represent better performance.

The countries contributing the most to EU27 patent and trademark output are Germany, France, and Italy, which altogether produced more than 60% of the patent and the trademark applications of the EU27 in 2023. On average, per capita patent and trademark output in EU MSs has increased by 17% and 6.5% respectively since 2013. The patenting activity of EU27 has remained relatively stable at around 100 patents per million population, lagging behind its main competitions such as the US, South Korea, and Japan (more than 150 patents per million population in 2023). In terms of trademarks, EU27 outperforms competitors such as the US and Japan, but still lags behind South Korea and China [3].

Figure 3 – Number of trademark classes of resident applications per million population (equivalent count), 2022



Source: Bello et al. 2025. Note: dotted line shows the EU27 average. Note: Data not available for Greece and Ireland.

In 2022, the EU27's exports to third countries were largely driven by knowledge-based products, with

approximately half of total good exports and over 80% of service exports falling into this category. A significant proportion of the value of these exports was generated within the EU itself. Notably, more than half of the value of the EU's medium-high and high-tech manufacturing exports originated from within the EU, on par with countries like South Korea and the US. The EU's service exports also demonstrated a substantial domestic value added, with 40% of their value created within the EU. While this is higher than many other countries, including China, Japan, and Australia, some countries like the US, Israel, and the UK still have a higher domestic contribution. Overall, this highlights the EU's strong domestic production capabilities and its competitive position in global trade.

Addressing the strategic dependencies along key supply chains remains important to guarantee the ability of the EU to pursue its economic security and resilience. On average, EU MSs contributed around one third to the value of their medium-high and high-tech manufacturing exports and their knowledge-intensive service exports in 2022. These figures have shown only a slight increase over the last decade. Notably, Germany is the EU MS that is least reliant on foreign inputs for its knowledge-based export products, while Finland adds the most domestic value to its service exports (Figure 4).

Employment in knowledge-intensive sectors remains low in the EU, with the share of employment in knowledge-intensive activities and industries currently standing at around 15%. This is comparable to Japan (16%), but well below that of other countries, like the United Kingdom (21%) and the United States (19%). The top performers among EU MSs are Luxembourg (28%) and Ireland (23%).

Nonetheless, knowledge-intensive sectors of the economy do not always dominate innovation activities. Firms in industries such as agriculture and mining also engage in innovative practices, introducing new products to the market, adopting novel business processes (e.g., sustainable ones), and conducting the necessary research and development (R&D) to support these efforts. The IOI's final component, the share of innovative enterprises, provides a comprehensive measure of innovation activity addressing both sectoral upgrading and structural changes.

Box 1: Inputs to innovation and the IOI

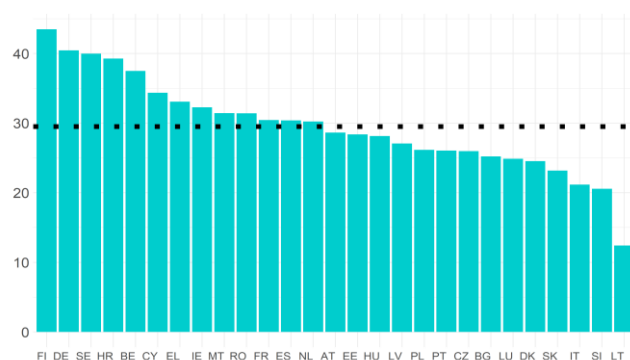
Innovation outputs are the result of the effective functioning of national innovation ecosystems.

Most countries that perform well on total innovation output – as captured by the IOI – are also active spenders in R&D. However, differences in R&D spending only partly explain differences in innovation outputs. For example, South Korea's IOI score is comparable to those of the US and Germany (133% vs. 134% and 130% respectively [3])², but its gross R&D expenditure as a share of GDP is significantly higher (5.2% vs 3.6% and 3.1% respectively [5]). At the same time, Switzerland outperforms the US and Germany in terms of the IOI [3] while spending roughly the same in R&D (3.3% of its GDP in 2021). This highlights that, while R&D spending is essential for fostering innovation, it is not the sole driver.

A comprehensive list of innovation “inputs” is provided by the European Commission's European Innovation Scoreboard (EIS) [4] in its “Framework conditions” and “Investment” pillars. These inputs monitor a multitude of factors ranging from the share of population involved in lifelong learning to venture capital expenditures and ICT specialists in employment. Countries' innovation outputs –as summarised in the IOI– are in close concordance with their innovation inputs as summarised in the EIS. Indeed, the IOI is well correlated with both the “Framework conditions” and “Investment” pillars and their component indicators. Switzerland, for example, that tops the IOI ranking, provides the best observed set of “Framework conditions” for innovation, while it is among the top countries in terms of “Investments” [4].

Another monitoring framework, WIPO's Global Innovation Index (GII) [6] aligns closely with these findings. Switzerland is ranked at the top of the GII for 14 consecutive years, while it is currently the second best country in terms of the GII “Inputs” pillar.

Figure 4 – Domestic value added content of exports in knowledge-intensive service industries as percentage of total service exports, 2020



Source: Bello et al. 2025. Note: dotted line shows the EU27 average.

According to the 2024 IOI results, more than half of all firms in the EU engage in innovation activities, a proportion similar to that of Japan but still lower than that of the US. However, a cause for concern is that the EU's performance has remained relatively stable since 2013. This overall figure conceals significant variations across EU Member States [3]. Notably, during the same period, the share of innovative enterprises declined by more than 10 percentage points (pp) in countries such as Germany, Luxembourg, and Romania, whereas it increased by more than 10 pp in countries like Cyprus, Greece, Lithuania, Croatia, Finland, and Belgium.

The stagnation in the innovation activities of EU firms is to be addressed through a series of forthcoming initiatives to be launched by the European Commission. Closing the innovation gap constitutes one of the main pillars of the Competitiveness Compass. Efforts to mitigate this gap will be supported through a dedicated EU Startup and Scaleup Strategy, aimed at fostering a favourable environment for young companies to launch, grow and scale up. Enterprises seeking to adopt advanced technologies, such as artificial intelligence (AI) and robotics, will benefit from the “Apply AI” initiative. In addition, a European Innovation Act will be proposed to, among other objectives, facilitate access to venture capital and support the testing of new solutions and technologies.

² where 100 is the EU27 value in 2013.

Conclusions and policy recommendations

This policy brief examines the innovation performance of the EU, building on the findings of the European Commission's Innovation Output Indicator (IOI). The index, which has been produced since 2013, tracks the extent to which innovative ideas reach the market, create better jobs, and increase a country's technological capability.

Despite progress, Europe still lags behind its main competitors in terms of innovation output. To close this gap, the EU should address its persistent disparities in research and innovation performance across Member States and tackle its insufficient private and public investment in R&D [1, 3, 5], as part of a more comprehensive strategy to boost innovation. This also includes promoting entrepreneurship, improving education and training, and creating a supportive business environment. In 2023, the EU spent 2.1% of its GDP on R&D, which is much less than South Korea (5.0%), the US (3.5%), Japan (3.4%), and China (2.6%) [7]. This shortfall is particularly notable given the EU's goal of investing 3% of its GDP in R&D, a target that remains elusive.

As highlighted by Draghi in his report, the EU should also rebalance its R&D and innovation efforts towards advanced technologies, particularly in areas where it evolves slower than its competitors. Currently, EU R&D investment and patenting activity are concentrated on mature technologies and sectors with stagnating productivity growth and moderate R&D intensity. Notably, the EU trails behind its main competitors in digital technologies, a critical area for future growth and competitiveness. Compared to the US and China, the EU is, for instance, less specialised in key productivity-enhancing technologies, particularly in the fields of artificial intelligence, internet of things, blockchain technologies and quantum computers [5]. In contrast, countries like South Korea are making significant strides in cutting-edge technologies, such as semiconductors, 6G, and ICT infrastructure, and are actively promoting their biotech industries [9].

The EU also requires a streamlined regulatory framework that facilitates business activities across the Union as a whole [1]. By eliminating regulatory barriers and inconsistencies across national borders,

Box 2: The case of Lithuania

Lithuania is the EU Member State that has made the most significant improvement in the IOI, with progress recorded across nearly all IOI dimensions. The country actively promotes the commercialization of research through a network of nine science parks and local technology centres. A notable example is the Photovoltaic Technology Cluster, which was established in Vilnius in 2008 and has since developed into one of the strongest national photovoltaic (PV) sectors in the EU. Many of its companies are now recognized as major players at the EU level [10].

At the regional level, Vilnius is considered one of Europe's most attractive investment destinations, drawing in high-tech companies with significant capital investments. The city boasts a well-developed infrastructure for high-tech manufacturing companies and is oriented towards innovative business practices, such as Fintech. Additionally, the country has a relatively efficient public agency, Invest Lithuania, which is responsible for attracting foreign direct investment (FDI) and providing support to investors [11].

In 2012 Lithuania, in collaboration with Estonia, Latvia and the European Investment Fund (EIF), set up a "hybrid" venture capital programme called the Baltic Innovation Fund (BIF)³, with the objective of boosting equity investment in Baltic small and medium-sized enterprises (SMEs) with high growth and international development potential [10]. By 2020, the BIF had supported investments totalling EUR 522 million in the three countries, comprising EUR 130 million in direct investments from the EIF and the countries' national agencies, and the remainder consisting of co-investments from private investors and pension funds. The second iteration of the programme, BIF 2, was launched in 2019 and aims to provide over EUR 500 million in new financing to Baltic companies.

the EU can reduce unnecessary costs and complexities for businesses. Moreover, simplifying rules, reducing the regulatory burden, and streamlining administrative procedures are essential

³ https://www.eif.org/what_we_do/resources/BIF/index.htm.

to create a more attractive and business-friendly environment, conducive to investment and growth.

All these shortcomings and areas of improvement are addressed in the Commission's communication on the Competitiveness Compass. This compass provides a comprehensive roadmap for revitalising Europe's economic growth and prosperity, with 'Closing the innovation gap' Pillar at its core. This suite of initiatives aims to reignite the EU's innovation engine by creating a supportive ecosystem for young start-ups, promoting industrial leadership in high-growth sectors driven by deep technologies, and facilitating the diffusion of technologies across established companies and SMEs. To achieve this, the Commission has proposed a dedicated EU Startup and Scaleup Strategy and a new Single Market Strategy, which aim to address the obstacles hindering the emergence and growth of new companies. Similarly, the forthcoming European Innovation Act will facilitate access for innovative companies to European research and technology infrastructures, as well as to intellectual assets arising from publicly funded R&I, with the aim of increasing patenting activity [12]. It will also support the use of regulatory sandboxes, enabling innovators to develop and test new ideas in a controlled environment.

To target the absence of a unified regulatory framework across the EU27, the Commission will propose a 28th legal regime. This regime will simplify applicable rules and reduce the costs of failure, making it easier for businesses to operate across the EU. The ERA Act, a flagship initiative under the Competitiveness Compass, will also play a crucial role in strengthening R&D investment, prioritising strategic research areas, and enhancing collaboration between the EU and its Member States, while promoting the free circulation of knowledge and talent across Europe, ultimately driving innovation and competitiveness.

The EU possesses the talent and capabilities needed to be an innovation leader [1, 4, 5]. However, targeted policy actions are essential to unlock the EU's full potential by boosting R&D investment, refocusing efforts on advanced technologies, and cultivating a supportive ecosystem for start-ups. The Competitiveness Compass, along with its associated Acts and initiatives, offers a comprehensive roadmap for achieving these objectives: fostering growth, enhancing competitiveness, generating employment, and positioning the EU as a global frontrunner in research and innovation.

Related work

This work is a deliverable of the INNOVA MEASURE VI project ([JRC 36562]). More information can be found on: https://knowledge4policy.ec.europa.eu/projects-activities/in-nova-project_en

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